

Case Study 1: Volkswagen Emissions Scandal

The Volkswagen emissions scandal illustrates how business objectives can conflict with ethical responsibilities. Volkswagen installed software, commonly referred to as a "defeat device" in certain diesel vehicles. This software detected when a vehicle was undergoing an emissions test and temporarily altered its performance so that it appeared to meet environmental standards. Under normal driving conditions, however, the vehicles emitted significantly higher levels of pollutants. The software was therefore intentionally designed to mislead regulators and consumers.

Q1: Which ethical theory aligns with Volkswagen's actions?

The ethical theory that most closely aligns with Volkswagen's conduct is ethical egoism. Ethical egoism holds that an action is morally acceptable when it advances the interests of the individual or organization making the decision. From Volkswagen's standpoint, the software supported key business objectives. It enabled the company to market diesel vehicles as environmentally friendly while preserving performance and remaining competitive with other manufacturers. Volkswagen also faced considerable pressure to produce a "clean" diesel vehicle for the American market, and the engineers involved were reportedly under intense internal pressure to meet this goal. From an ethical-egoist perspective, the company may have prioritized its business objectives, market position, and profits over the negative consequences for consumers, regulators, and the environment.

It is important to note, however, that this alignment does not justify Volkswagen's actions. It merely explains how the decisions could have been rationalized from a self-interest-based perspective.

Q2: Which ethical theory could refute Volkswagen's actions?

Utilitarianism offers a strong refutation of Volkswagen's conduct. Utilitarianism evaluates an action by its overall consequences and seeks the greatest benefit for the greatest number of people. Volkswagen's decision produced short-term benefits for the company, but the negative consequences affected a far larger group. Consumers were deceived into purchasing vehicles under false environmental claims, regulators were deliberately misled, competitors were placed at a disadvantage, and excessive emissions contributed to environmental and public-health problems. The scandal also damaged Volkswagen's reputation and diminished public trust.

A deontological approach would likewise reject Volkswagen's behavior, as deliberately deceiving consumers and regulators is wrong regardless of any financial benefits obtained. The software engineers, therefore, had a responsibility not to develop a system designed to deceive.

Overall, the Volkswagen case demonstrates that business efficiency and competitive pressure cannot justify unethical software development. Software engineers have a responsibility to consider how their work affects society, not merely whether it achieves the company's immediate objectives.

Case Study 2: Boeing 737 MAX Disasters

The Boeing 737 MAX disasters illustrate the severe consequences that arise when business competition, cost reduction, and engineering decisions are prioritized over safety. Two Boeing 737 MAX aircraft crashed in 2018 and 2019, resulting in the deaths of 346 passengers and crew. Investigations and ethical analyses raised concerns regarding the aircraft's design, the MCAS software, pilot training, and organizational decision-making.

Q1: Which ethical theory aligns with Boeing's actions?

The ethical theory that most closely explains Boeing's decisions is ethical egoism. Ethical egoism holds that an action is morally acceptable when it advances the interests of the individual or organization making the decision. Boeing was engaged in intense competition with Airbus and sought to keep the 737 MAX competitive without requiring major design changes or costly additional pilot training. The larger engines altered the aircraft's aerodynamics, and Boeing introduced the MCAS software to compensate for these changes.

From an ethical-egoist perspective, Boeing's decisions can be interpreted as attempts to protect its own interests, including reducing costs, maintaining market position, and delivering the aircraft quickly. Avoiding additional training and certification requirements could make the aircraft more appealing to airlines and lower expenses.

The difficulty is that this approach placed corporate interests above the safety of passengers, pilots, and the public. As the APA article argues, Boeing's decisions conflicted with the engineering principle that the safety, health, and welfare of the public must be treated as paramount.

Q2: Which ethical theory could refute Boeing's actions?

Utilitarianism provides a strong argument against Boeing's decisions because the consequences were extremely harmful. Even if Boeing's choices produced short-term economic benefits, those gains were insignificant compared with the loss of human life and the broader damage caused by the crashes.

A deontological approach offers an even stronger criticism. Deontology emphasizes duties and moral rules rather than merely the consequences of an action. Engineers and managers working on safety-critical aircraft have a duty to protect passengers and to provide pilots with the information necessary to operate the aircraft safely. The APA analysis specifically discusses concerns that pilots were not properly informed about the new automatic anti-stall system and emphasizes the precautionary principle when dealing with uncertain safety risks.

The case also demonstrates why reducing testing or concealing important information cannot be justified simply because a company wishes to remain competitive. When human lives depend on software, safety must take priority over cost, speed, and profit. The Boeing case therefore shows that ethical engineering requires professionals to place public safety ahead of organizational self-interest.

References

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